

FedID: Enhancing Federated Learning Security through Dynamic Identification

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Abstract—Federated learning (FL), recognized for its decentralized and privacy-preserving nature, faces vulnerabilities to backdoor attacks that aim to manipulate the model's behavior on attacker-chosen inputs. Most existing defenses based on statistical differences take effect only against specific attacks. This limitation becomes significantly pronounced when malicious gradients closely resemble benign ones or the data exhibits non-IID characteristics, making the defenses ineffective against stealthy attacks. This paper revisits distance-based defense methods and uncovers two critical insights: First, Euclidean distance becomes meaningless in high dimensions. Second, a single metric cannot identify malicious gradients with diverse characteristics. As a remedy, we propose FedID, a simple yet effective strategy employing multiple metrics with dynamic weighting for adaptive backdoor detection. Besides, we present a modified z-score approach to select the gradients for aggregation. Notably, FedID does not rely on predefined assumptions about attack settings or data distributions and minimally impacts benign performance. We conduct extensive experiments on various datasets and attack scenarios to assess its effectiveness. FedID consistently outperforms previous defenses, particularly excelling in challenging Edge-case PGD scenarios. Our experiments highlight its robustness against adaptive attacks tailored to break the proposed defense and adaptability to a wide range of non-IID data distributions without compromising benign performance.

Index Terms—Federated learning, Backdoor attack, Robust defense, Safe aggregation.

I. INTRODUCTION

FEDERATED learning (FL) [1], [2] represents a distributed machine learning paradigm that facilitates collaborative model training among multiple participants, all while preserving the privacy of their local data. In each training round, a central server distributes the global model to a selected subset of clients, who then update the model with their respective local data and transmit the parameters back to the server for aggregation. FL offers an effective approach to training efficient, high-quality models in a decentralized manner, all without compromising the privacy of the participating entities.

This work is supported by the National Natural Science Foundation of China (Grant No. 62476095).

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This characteristic makes FL a suitable solution for reconciling the potential conflicts between technological advancements and stringent data privacy regulations, such as the General Data Protection Regulation (GDPR). The applicability and success of FL span across various domains, including but not limited to image processing [3], word prediction [4], medical imaging [5], and edge computing [6], [7].

However, FL faces vulnerabilities to backdoor attacks that manipulate the model towards specific targeted behaviors using adversary-chosen inputs [8]–[15]. Detecting backdoor attacks is more challenging compared to untargeted poisoning attacks [10], [14], [16]–[18] because they do not disrupt the model's regular functioning, and their gradients closely resemble benign ones [11], [19]. Various defenses have been proposed to enhance FL's robustness, including distance-based methods [20]–[30], which rely on specific metrics to differentiate malicious gradients from benign ones. However, researchers have discovered that well-designed attacks, known as *stealthy backdoors*, can bypass these defenses by making their gradients indistinguishable from benign ones through techniques like scaling [9], [11] or trigger splitting [12]. Differential privacy (DP)-based methods [19], [31]–[34] have also been developed, leveraging the effectiveness of the DP method [35] against backdoors. These methods add Gaussian noise to the global model to mitigate the impact of potentially poisoned model updates, showing significant resilience against *stealthy backdoors* like Edge-case PGD [11]. While DP-based methods can resist *stealthy backdoors*, the added noise significantly hinders overall performance and convergence speed. In contrast, distance-based methods aggregate only benign gradients, raising a fundamental question: *Can we defend against stealthy backdoors without sacrificing FL model performance?* To address this question, we turn to distance-based methods that maintain benign performance and explore how to effectively employ distance metrics to discriminate hostile updates from benign ones.

To this end, we revisit the distance-based defense and discover two key limitations: 1) *Challenges in High-Dimensional Space*: The Euclidean distance (i.e., L_2 distance) is hindered by the curse of dimensionality. In the context of neural networks (NNs), where parameters can be represented as high-dimensional vectors, Euclidean distance often fails to effectively distinguish between malicious and benign gradients in these expansive, high-dimensional spaces. 2) *Limitations of Single Metric Approaches*: The efficacy of a single metric is limited, as it only proves effective against specific attacks that rely on detailed assumptions about the characteristics of

malicious gradients. For instance, the cosine distance metric is adept at detecting malicious gradients with substantial angular deviations, while the Euclidean distance metric excels at identifying malicious gradients with a large L_2 norm that attackers have scaled to influence global models. Moreover, backdoor attacks are conducted under varying data and scenario conditions, resulting in malicious gradients that exhibit diverse characteristics. In a single round of aggregation, one may encounter gradients with markedly large norms and those with significant angular deviations. This diversity presents a formidable challenge, as a single metric is ill-suited to address this complexity. What's worse is that the defender does not know the attacker and the underlying data distributions due to privacy constraints imposed by FL, which makes detecting with a single metric even more difficult.

To tackle the abovementioned issues, we first introduce the Manhattan distance, which we theoretically demonstrate to be more meaningful in high-dimensional spaces than the Euclidean distance. Empirically, our proposed distance-based defense, when using the Manhattan distance, exhibits remarkable performance against *stealthy backdoors*. In order to address gradients with diverse properties, we cooperatively employ multiple metrics to identify malicious gradients. In our ablation study, we provide empirical evidence that specific metrics excel at identifying malicious gradients with particular characteristics, justifying the motivation of our approach. To handle the different attacks and environments, we further propose to apply a whitening transformation and generate dynamic weights to handle the non-IID distribution of participants and different scales brought by the different distances. After calculating the score for each submitted gradient, we introduce a modified z-score approach for anomaly detection and the dynamic selection of updates to be aggregated. Moreover, we posit that potential attacks may attempt to conceal themselves to evade our defense adaptively. However, our experiments reveal that such potential adaptive attacks are either detected or rendered non-threatening to the FL system. Our extensive experiments illustrate that our defense successfully maintains both high model performance and robustness simultaneously, achieving a feat unprecedented in prior research. Comprehensively, we summarize the main contributions of this paper as follows:

- We address the “meaningfulness” challenge of Euclidean distance in high-dimensional spaces by incorporating the Manhattan distance into our defense and demonstrate its superiority in theory and experiment.
- We utilize multiple metrics with dynamic weighting, enabling our defense to effectively withstand backdoor attacks across various attack scenarios and data distributions.
- We devise a modified z-score method for anomaly detection, which adapts to our approach by dynamically selecting updates for aggregation based on their distance scores.
- Our work is the first defense capable of effectively countering *stealthy backdoors* without the need for predefined assumptions or compromising the main performance and even countering potential adaptive attacks targeted at the

defense.

Compared with our original conference version [36], the following improvements are conducted: 1) We further enhance our method by devising a dynamic anomaly detection approach, replacing a fixed parameter to decide the portion to aggregation; 2) We present the potential adaptive attack targeted to our approach from the perspective of the full-knowledge adversary and prove the robustness of FedID; 3) We employ experiments on more extensive datasets (e.g., Fashion-MNIST) and attack scenarios (e.g., untargeted attacks) to verify the effectiveness, generalization ability, and scalability of our FedID.

II. RELATED WORK

A. Backdoor Attacks on Federated Learning

FL enables multiple clients to collaboratively train a model without local data exchange under the orchestration of the server. In an FL system, clients update the local model with private data and submit the parameters to the server for aggregation. Given a total of N participants, each with $n^{(i)}$ samples. In each round t , the server randomly selects K clients to participate. FedAvg [2] is a widely adopted FL algorithm, which can be formulated as:

$$w_t = w_{t-1} + \eta \cdot \frac{\sum_{i=1}^K n^{(i)} \Delta w_t^{(i)}}{\sum_{i=1}^K n^{(i)}} \quad (1)$$

where w_t is the global model parameters at round t , $\Delta w_t^{(i)}$ is the local model updates from the client i in round t , and η is the global model learning rate.

Empirical studies conducted by Bagdasaryan et al. [9] have demonstrated that FedAvg [2] is susceptible to backdoor attacks, primarily due to FL's lack of control over local data and its training process, driven by stringent privacy concerns. However, due to the overwhelming updates from numerous benign clients, the injected backdoors can easily be diluted [8], [9]. To inject backdoors into the FL model effectively, various strategies have been devised. One such strategy is the model replacement attack [9], which involves amplifying the attack gradients and successfully injecting a backdoor into the global model via a single-shot attack. These scaling-up methods, however, are often straightforward and can be readily detected by existing defenses [37]. To circumvent existing defenses, attackers have developed more sophisticated approaches, such as the PGD attack [11], which involves scaling and projecting gradients, and the DBA attack [12], where triggers are split before being uploaded. Additionally, the Edge-case attack [11] leverages poisoned data located in the long tails of data distributions, areas unseen in benign client datasets. This approach ensures that the backdoors remain undiluted and effective. Furthermore, the Edge-case PGD attack [11] goes a step further by modifying both the poisoned data and the model, enhancing its effectiveness and stealthiness. Undoubtedly, these attacks present significant challenges to the security of FL systems.

B. Defense Against the Backdoor Attack.

Various defense methods have been developed to enhance the robustness and security of FL against backdoor attacks. Broadly, these methods can be categorized into two types.

The first type of defense strategy aims to discriminate between malicious gradients and benign ones by classifying or scoring them under the assumption that benign and malicious gradients exhibit distinct differences in the vector space [20]–[29]. For example, Krum and Multi-Krum [21] select the gradient with the smallest Euclidean norm from the others for aggregation and updating of the global model in each round. Foolsgold [20] operates under the assumption that the gradients of attackers consistently exhibit specific patterns and distinguishes them using the Cosine similarity between historical updates. Trimmed Mean and Median [22] remove a portion of gradients with maximum and minimum values, then compute the median value for each dimension of the remaining gradients to derive the global model's parameters. RFA [23], on the other hand, employs the geometric median of gradients to maintain proximity to the benign model while distancing itself from the backdoor-infected model. The strength of these approaches lies in their ability, if successful, to exclude attacker gradients from the aggregation process, effectively addressing the core and fundamental challenge of defense against such attacks.

However, the aforementioned methods are effective only in specific attack scenarios, as they rely on detailed assumptions about the attack or the underlying data distributions. For instance, Foolsgold [20] assumes that attacker gradients consistently exhibit a specific pattern, while benign gradients from non-IID data differ. Krum [21], on the other hand, assumes that the data from benign clients are similar under IID settings and excludes outliers that are deemed malicious. These predefined assumptions restrict the applicability of these defenses in the face of diverse attacks and varying data distributions, making them susceptible to sophisticated attacks. For example, the PGD attack [11] can effectively bypass any Euclidean-based defense, including methods like Krum and Multi-Krum, by scaling the gradient norm to a small value. Other methods, such as FLTrust [24], rely on a root dataset and aggregate gradients that exhibit large cosine similarities with this root dataset. FedInv [25] takes the approach of inverting gradients into dummy datasets and then identifies malicious gradients by comparing them with these dummy datasets. However, these methods have limitations of their own. FLTrust cannot defend against backdoors specifically crafted for edge cases (e.g., Edge-case PGD) since such cases occur infrequently and may not be present in the root dataset. Inverting gradients into dummy datasets violates the privacy requirements of FL and can essentially be categorized as another form of attack [38]–[41].

The other category of defense methods [19], [31]–[34] is based on the recognition that the differential privacy (DP) method [35], originally designed to counter DP attacks, can also effectively combat backdoor attacks. Prior research [9] has demonstrated that techniques such as clipping and noise addition, commonly employed in DP, can successfully mitigate

backdoor attacks without relying on specific assumptions about data distribution. Building on these insights, Weak-DP [31] utilizes gradient norm clipping and Gaussian noise addition to the global model, effectively mitigating backdoor threats. Meanwhile, RLR [32] and Flame [19] incorporate clustering methods into the DP process.

Nevertheless, it's worth noting that the introduction of noise significantly impacts the performance and convergence speed of the FL system. In essence, using DP sacrifices regular functionality in favor of enhanced robustness. Despite its drawbacks, the DP-based method remains the only defense that has proven effective against *stealthy backdoors* [19].

C. Threat Model

Adversarial goals. At a high level, the adversary aims to alter the model's behavior on some specific examples with no negative impact on the benign performance. The goals of the adversary are two-fold: 1) Effectiveness: the poisoned model predicts incorrect labels on adversary-chosen samples, which usually with pixel-trigger [12], [14] or semantic-trigger [9], [11]. Importantly, these poisoned samples are only in the dataset of the adversary. 2) Stealthiness: the modified model behaves the correct prediction on benign samples, which belong to datasets of both benign clients and the adversary. We consider the adversary effective if it satisfies both of these criteria, effectiveness and stealthiness. In addition, it is worth noting that a simple data-poisoned attack or straight scaling appears to be naive to success against the range of FL defense [37]. Consequently, the adversary is compelled to employ sophisticated, covert strategies in their attacks [9], [11], [12]. Formally, adversary $\mathcal{A}$ manipulates the global model G to the poisoned model G' . For any inputs x , we have the outputs below:

$$G'(x) = \begin{cases} l' \neq G(x) & \forall x \in D_{\mathcal{A}} \\ G(x) & \forall x \notin D_{\mathcal{A}} \end{cases} \quad (2)$$

where $D_{\mathcal{A}}$ denotes the poisoned dataset belonging to adversary $\mathcal{A}$, and l' denotes the specific label modified by adversary $\mathcal{A}$ of the input x .

Adversarial capabilities. In this paper, we operate under the white-box setting, granting adversaries full knowledge of the aggregation operations. This includes insights into potentially deployed defenses, aggregation algorithms, and configuration parameters. Additionally, adversaries have the freedom to modify parameters in local training, ensure their participation in each round, and compromise fewer than 50% of the total FL clients. It's crucial to emphasize that the adversary lacks control over benign clients, access to their local data or models, and cannot tamper with any process or parameter executed within the aggregation.

III. METHODOLOGY

We first provide the threat model in Section II-C and then detail our design of the proposed defense method. Our goal is to design an efficient defense with the following properties: (1) Effectiveness: To prevent the adversary from achieving its hostile purpose, we aim to design a defense that identifies as

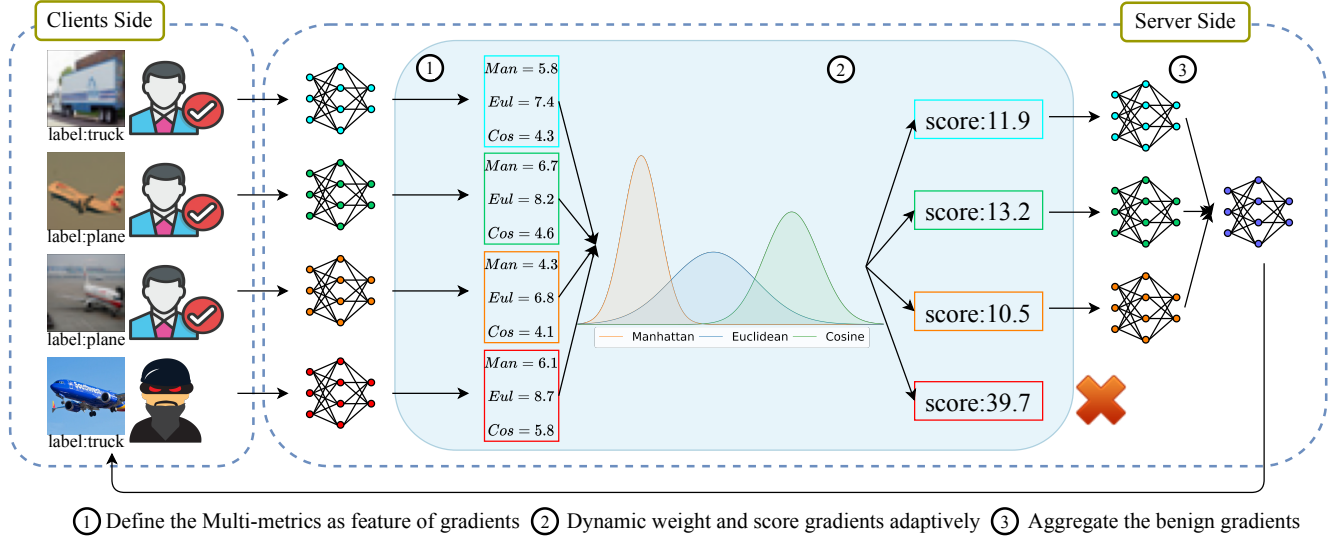


Fig. 1. The overview of the FedID process. Step 1 and step 2 are our core contributions.

many malicious updates as possible to eliminate its impact on the global model. (2) Performance: The benign performance of the global model must be preserved to maintain its utility. (3) Independence: The defense must apply to generic adversary models under different data distributions and attack strategies. Our defense FedID consists of three major components: defining and calculating the gradient feature, computing the dynamic weight and score for each update, and aggregating the benign gradients with the score computed, as shown in Figure 1 and Algorithm 1.

A. Feature of gradient

The essential logic of distance-based defense involves defining some indicative metric that can well discriminate the malicious gradients from the benign ones and remove the hostile updates from the aggregation. Consequently, the core problem becomes how to define a metric that identifies the characteristics of hostile gradients. For instance, NDC [31] utilizes the length of the gradient (i.e., Euclidean distance) while FoolGold [20] leverages Cosine similarity to measure the difference between updates.

Manhattan Distance. However, the Euclidean distance suffers from the so-called “curse of dimensionality” [42], which renders distance metrics less sensitive in high dimensional space, especially in the case of Euclidean distance. Theorem 1 uses $\frac{Dmax_d^k - Dmin_d^k}{Dmin_d^k}$ to indicate the “meaningfulness” of a distance referred as the *Relative Contrast*. It demonstrates that the distance metric is meaningless in high-dimensional space because as dimensionality increases, the distance between the nearest and the furthest neighbor approximates to be the same.

Theorem 1 (Beyer et. al. [43] The curse of dimensionality):

$$\text{If } \lim_{d \rightarrow \infty} \text{var} \left(\frac{\|X_d\|_k}{E[\|X_d\|_k]} \right) = 0, \text{ then} \\ \frac{Dmax_d^k - Dmin_d^k}{Dmin_d^k} \rightarrow_p 0,$$

where d denotes the dimensionality of the data space, $E[X]$ and $\text{var}[X]$ denotes the expected value and variance of a random variable X , $Dmax_d^k$ and $Dmin_d^k$ means the farthest/nearest distance of the N points to the origin using the distance metric L_k .

To find out which distance metric is more meaningful in high-dimensional space, we size up the relation between the dimension d and the distance L_k . We have Lemma 1, which shows that $Dmax_d^k - Dmin_d^k$ increases at the ratio of $d^{(1/k)-(1/2)}$.

Lemma 1 (Hinneburg et.al. [44]): Let $\mathcal{F}$ be an arbitrary distribution of two points and the distance function $\|\cdot\|$ be an L_k metric. Then,

$$\lim_{d \rightarrow \infty} E \left[\frac{Dmax_d^k - Dmin_d^k}{d^{(1/k)-(1/2)}} \right] = C_k,$$

where C_k is some constant dependent on k .

Subsequently, we use the value of $Dmax_d^k - Dmin_d^k$ as a criterion to evaluate the ability of a particular metric to discriminate different vectors in high dimensional space.

Proposition 1: Let $M_d = Dmax_d^1 - Dmin_d^1$ reflect the discriminating ability of Manhattan distance and $U_d = Dmax_d^2 - Dmin_d^2$ reflect the discriminating ability of Euclidean distance. Then,

$$\lim_{d \rightarrow \infty} E \left[\frac{M_d}{U_d \cdot d^{\frac{1}{2}}} \right] = C'.$$

where C' is a constant. *Proof* see in Appendix A.

Proposition 1 shows that M_d and $U_d \cdot d^{\frac{1}{2}}$ are infinite of the same order. Since $d^{\frac{1}{2}}$ is also an infinite value, M_d towers above U_d , suggesting that Manhattan can discriminate more than the Euclidean distance in high-dimension space. Given this, we conclude that the Manhattan distance is better than the Euclidean metric, withstanding the curse of dimensionality, which chimes in with the previous experiment [45]–[47]. The parameters of neural networks can be viewed as a typical high-dimensional space, especially when the number of parameters

increases as the trend in deep learning applications [48]–[53]. In light of this and Theorem 1, we argue that Euclidean distance is insufficient to discriminate between malicious and benign gradients. With Proposition 1, we introduce the Manhattan distance metric to describe the characteristics of gradients, which captures the difference in high dimensional space better.

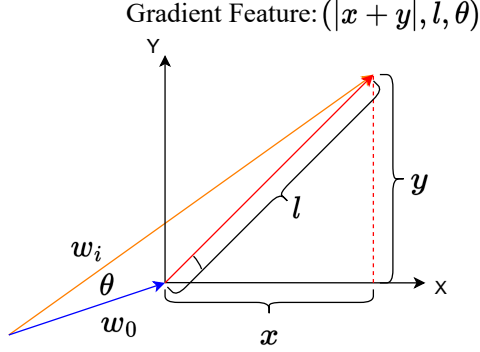


Fig. 2. Demonstration of *feature of gradient* on two dimensional space. w_0 is initial global model w_0 and w_i is the local model of the i -th client. $w_i - w_0$ is the gradient and we define the *feature of gradient* as $(|x + y|, l, \theta)$.

Multiple Metrics as Feature of Gradient. Another problem that we identify as discussed in Section I is that current methods conduct defense on a single metric basis. With only one metric, a sophisticated attacker could effortlessly bypass them with a well-designed gradient. For instance, PGD attack [11] manages to break the defense based on Euclidean distances by clipping the gradient norm to a small value. Moreover, attackers conduct attacks under different environments and data distributions, leading to malicious gradients with diverse characteristics that a single metric cannot handle. To this end, we propose multiple metrics to cooperatively identify the malicious gradient by defining the *feature of gradient* as the angle by the Cosine similarity, length by the Euclidean distances, and its Manhattan norm by the Manhattan distance, namely, $\mathbf{x} = (x_{Man}, x_{Eul}, x_{Cosine})$. Here, x_{Man} denotes the Manhattan distance, x_{Eul} denotes the Euclidean distance, and x_{Cosine} denotes the Cosine similarity. Figure 2 provides a schematic diagram of the *gradient feature*. We compute the gradient feature x_{Man} , x_{Eul} , and x_{Cosine} of the i -th client of each round as follows: $x_{Man}^{(i)} = \|w_i - w_0\|_1$, $x_{Eul}^{(i)} = \|w_i - w_0\|_2$, $x_{Cosine}^{(i)} = \frac{\langle w_0, w_i \rangle}{\|w_0\| \cdot \|w_i\|}$, where w_0 denotes the global model before federated training and w_i denotes the i -th client's local model after training, $\|\cdot\|_1$ means the L_1 norm of a vector, $\|\cdot\|_2$ means the L_2 norm of a vector, $\langle \cdot, \cdot \rangle$ and represents inner product.

After defining the *feature of gradient*, we utilize it for malicious identification. The goal is to identify the outlier among the gradients. We use the sum of the distance between each gradient as the indicator. Formally, we define it as Equation 3.

$$\mathbf{x}'^{(i)} = \left(\sum_{j,j \neq i}^K |x_{Man}^{(i)} - x_{Man}^{(j)}|, \sum_{j,j \neq i}^K |x_{Eul}^{(i)} - x_{Eul}^{(j)}|, \sum_{j,j \neq i}^K |x_{Cosine}^{(i)} - x_{Cosine}^{(j)}| \right), \quad (3)$$

B. Dynamic Weighting Through Whitening

After defining the *feature of gradient* and indicator, we now provide an approach to scoring each gradient. Two factors are taken into account when considering an efficient way to turn the indicator into a unified score that works under various data distributions and attacks.

- Different scales of the three distance metrics are the first obstacle before utilizing these metrics collaboratively. Since each metric is correlated, a novel regularization is required instead of the usual normalization by the maximum value, as shown in Fig3.
- Different data distributions (e.g. different degrees of non-IID) render the gradients of both malicious and benign clients different. Thus, dynamic weighting is required to cope with various environments and attacks to achieve a universal defense.

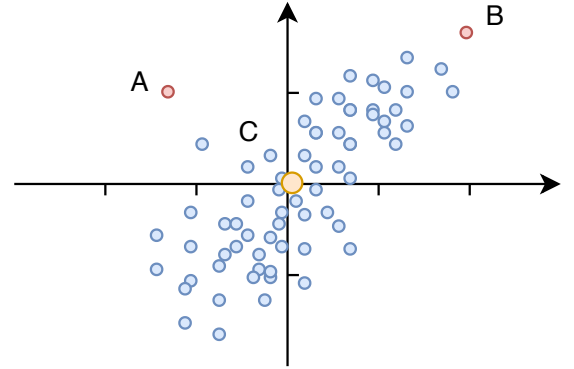


Fig. 3. Point C is the center point among all points. Although the distance between C and A is less than between B, we consider point A as the abnormal point.

Considering the above factors, we propose first to project the value of the metrics in the *feature of gradient* to its corresponding principal axis by applying a whitening process:

$$\delta^{(i)} = \sqrt{\mathbf{x}'^{(i)\top} \Sigma^{-1} \mathbf{x}'^{(i)}}. \quad (4)$$

where Σ is the the covariance matrix of $\mathbf{X} = [\mathbf{x}'^{(1)}, \mathbf{x}'^{(2)}, \dots, \mathbf{x}'^{(K)}]^\top$. Since we need to calculate the inverse of the covariance matrix Σ , the number of samples must be larger than the number of features, i.e., $K > 3$. Please notice that the inverse of the covariance matrix is calculated based on the selected gradients, which will change the weights of the features dynamically depending on the feature distribution, hence "dynamic weighting". Our method can better adapt to different environments and resist attacks with such a dynamic weight.

C. Aggregation of Benign Gradients

Once we obtain the score for each gradient, we proceed to select and aggregate benign gradients with low scores. A lower score indicates that the gradient is less divergent compared to others, which is considered benign. The choice of proportion selected as benign is difficult since the defendant is unaware of the percentage of malicious clients during each round of updates. A larger proportion raises the likelihood of choosing backdoor gradients, leading to the potential injection of backdoors. Conversely, a smaller proportion of aggregation heightens the impact of undiluted malicious gradients, undermining the defense strategies.

Therefore, we propose dynamic thresholding by modifying the z-score to distinguish benign gradients from malicious ones in one round of updates. The z-score [54] is a widely used metric in anomaly detection but may not be suitable for few-shot samples.

$$\gamma = \frac{x - \mu}{\sigma}, \quad (5)$$

where μ and σ denote the mean and the standard deviation of the samples. Traditionally, a value x is considered an outlier if $|\gamma| > 3$. We modify the calculation of the z-score to adapt it for detecting anomalies in the submitted gradients $[\delta^{(1)}, \delta^{(2)}, \dots, \delta^{(K)}]$ computed from Equation 4. Firstly, we employ the Median Absolute Deviation (MAD) to replace the σ , as MAD is more robust with few-shot samples and is less influenced by outliers.

$$\text{MAD}(\{\delta^{(i)}\}_{i=1}^K) = \text{median} \left(\left| \delta^{(i)} - \text{median}(\{\delta^{(i)}\}_{i=1}^K) \right| \right) \quad (6)$$

Subsequently, we use the minimum value of $[\delta^{(1)}, \delta^{(2)}, \dots, \delta^{(K)}]$ to replace μ since the minimum δ is assumed to be the most representative of the global gradient in our method. Formally we have,

$$\gamma^{(i)} = \frac{\delta^{(i)} - \min_i \delta^{(i)}}{\text{MAD}(\{\delta^{(i)}\}_{i=1}^K)}. \quad (7)$$

where $\gamma^{(i)}$ represents the degree of an outlier for the i -th gradient, and we consider gradients with $\gamma^{(i)} \geq 1$ as malicious and only aggregate the gradients with $\gamma^{(i)} < 1$. Once we have selected the ‘‘benign’’ gradients, we can aggregate them using any existing aggregation method (e.g., averaging). We follow the approach outlined in [2] and perform the standard FedAvg on the ‘‘benign’’ gradients.

IV. THEORETICAL ANALYSIS

A. Effectiveness Analysis

Firstly, we theoretically analyze the effectiveness of FedID in identifying malicious gradients. We illustrate that after computing the features of the gradients and then applying the whitening through dynamic weighting, the outlier score of benign gradients is smaller than malicious gradients as follows.

Theorem 2: Consider the feature of each gradient as a 3-dimension vector $v_i = (x_1, x_2, x_3)$. Given n vectors in a round, we denote the benign class as $\mathcal{B}$ and the malicious

Algorithm 1 FedID algorithm

Input: Total number of clients in each round K , models of clients $\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \dots, \mathbf{w}^{(K)}$, last global model $\mathbf{w}_0$, global learning rate η , size of the i -th client dataset $n^{(i)}$

Output: Global model $\mathbf{w}'$

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1: for  $i \in \{1, 2, \dots, K\}$  do ▷ compute the gradients
   features
2:    $x_{Man}^{(i)} \leftarrow \|\mathbf{w}_0 - \mathbf{w}_i\|_1$ 
3:    $x_{Eul}^{(i)} \leftarrow \|\mathbf{w}_0 - \mathbf{w}_i\|_2$ 
4:    $x_{Cos}^{(i)} \leftarrow \frac{\langle \mathbf{w}_0, \mathbf{w}_i \rangle}{\|\mathbf{w}_0\| \cdot \|\mathbf{w}_i\|}$ 
5:    $\mathbf{x}^{(i)} \leftarrow (x_{Man}^{(i)}, x_{Eul}^{(i)}, x_{Cos}^{(i)})$ 
6: end for
7: for  $i, j \in \{1, 2, \dots, K\}$  do ▷ compute the sum of the
   distance between each gradient
8:    $x'_{Man}^{(i)} \leftarrow \sum_{j, j \neq i}^K |x_{Man}^{(i)} - x_{Man}^{(j)}|$ 
9:    $x'_{Eul}^{(i)} \leftarrow \sum_{j, j \neq i}^K |x_{Eul}^{(i)} - x_{Eul}^{(j)}|$ 
10:   $x'_{Cos}^{(i)} \leftarrow \sum_{j, j \neq i}^K |x_{Cos}^{(i)} - x_{Cos}^{(j)}|$ 
11:   $\mathbf{x}'^{(i)} \leftarrow (x'_{Man}^{(i)}, x'_{Eul}^{(i)}, x'_{Cos}^{(i)})$ 
12: end for
13:  $X \leftarrow [\mathbf{x}'^{(1)}, \mathbf{x}'^{(2)}, \dots, \mathbf{x}'^{(K)}]^\top$ 
14:  $\Sigma \leftarrow$  the Covariance of  $X$  ▷ compute the adaptive
   weight matrix in this round
15: for  $i \in \{1, 2, \dots, K\}$  do
16:    $\delta^{(i)} \leftarrow \sqrt{\mathbf{x}'^{(i)\top} \cdot \Sigma^{-1} \cdot \mathbf{x}'^{(i)}}$  ▷ dynamically compute
   the divergency
17: end for
18: for  $i \in \{1, 2, \dots, K\}$  do
19:    $\gamma^{(i)} = \frac{\delta^{(i)} - \text{minimum}}{\text{MAD}}$  ▷ dynamically select gradients
   into aggregation
20:   if  $\gamma^{(i)} > 1$  then
21:     Remove this gradient from aggregation
22:   end if
23: end for
24: Set remaining gradients as  $\mathbb{B}$ 
25:  $\mathbf{w}' \leftarrow \mathbf{w}_0 + \eta \cdot \frac{\sum_{i \in \mathbb{B}} n^{(i)} \cdot (\mathbf{w}^{(i)} - \mathbf{w}_0)}{\sum_{i \in \mathbb{B}} n^{(i)}}$ 

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class as $\mathcal{M}$. We consider a challenging setting where the stealthy malicious gradient is similar to the benign ones and only has a distinction on x_1 . Let $v_{i \in \mathcal{B}} \sim \mathcal{N}(\boldsymbol{\mu}_B, \boldsymbol{\Sigma}_B)$, $\boldsymbol{\mu}_B = (\mu_{B1}, \mu_{B2}, \mu_{B3})$ and $\boldsymbol{\Sigma}_B = \text{diag}(\sigma_{B1}^2, \sigma_{B2}^2, \sigma_{B3}^2)$. For the gradients in $\mathcal{M}$, they follow the $\mathcal{B}$'s distribution in x_2 and x_3 and have different distributions in x_1 with different μ_{M1} and the same σ_{B1}^2 . Let the assumption of $|\mu_{B1} - \mu_{M1}| > \sigma_{B1}$ and the benign gradients be the majority, then we have the following promise of the outlier score $\delta^{(i)}$:

$$\mathbb{E}[\delta^{(i)}] < \mathbb{E}[\delta^{(j)}], \quad \forall i \in \mathcal{B}, \forall j \in \mathcal{M}$$

The detailed proofs are shown in Appendix B. This theorem ensures that the expected outlier score of malicious gradients is larger than that of benign ones. This finding shows that stealthy malicious gradients can be effectively identified by FedID, even though they have a slight deviation on just one feature. As $|\mu_B - \mu_M|$ increases, the malicious gradient is more easily detected. Therefore, our proposed approach can

thwart the attacks by subsequently selecting the gradients with minimum outlier scores to aggregate. However, there is also a probability of selecting the malicious gradients into aggregation, particularly in non-IID scenarios where the variance of benign gradients Σ_B is large. We further demonstrate the effectiveness against various attacks and the robustness in various scenarios of FedID in our experiments.

B. Convergence Analysis

Secondly, we establish the convergence of the FedID. First, we give several common assumptions on the function f , following the previous studies [55], [56].

Assumption 1 (L-smooth): For any sampled data z_i , the cost function $f(x; z_i)$ is L -smooth, i.e., it holds that $f(x; z_i) \leq f(y; z_i) + \langle \nabla f(y; z_i), y - x \rangle + \frac{L}{2} \|x - y\|^2$.

Assumption 2 (Bounded Variance): Let z_i be sampled from the i -th client's local data uniformly at random. We assume that the variance of gradients $\nabla f_i(w; z_i)$ is upper bounded by σ^2 , such that $\mathbb{E} \|\nabla f_i(w_t; z_i) - \nabla f_i(w_t)\|^2 \leq \sigma^2$, for all $i = 1, \dots, N$ and $t = 1, \dots, T$.

Assumption 3 (Bounded Gradient): We assume that the squared norm of stochastic gradients of the client is uniformly bounded, such that $\mathbb{E} \|\nabla f_i(w_t; z_i)\|^2 \leq G^2$, for all $i = 1, \dots, N$ and $t = 1, \dots, T$.

Assumption 4 (Bounded Malicious Impact): Let $\hat{g}_t$ be the ideal global gradient that only contains benign updates in $\mathcal{B}$, and $\bar{g}_t$ be the actual one that may contain malicious updates. We assume that the squared norm of the actual global gradient is uniformly bounded, such that $\mathbb{E} \|\bar{g}_t - \hat{g}_t\|^2 \leq \kappa^2$, for all $t = 1, \dots, T$, where $\hat{g}_t = \sum_{i \in \mathcal{B}} p_i \nabla f_i(w_{t-1}; z_i)$.

Then, in the following theorem, we provide the convergence analysis.

Theorem 3: Let Assumptions 1 – 4 hold, and L , σ , G , and κ be defined therein, and after T global rounds, then it has the following convergence guarantee of FedID:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(w_{t-1})\|^2] \leq \frac{2(f(w_1) - f(w^*))}{\eta T} + 2L\eta\Delta_1 + \Delta_2$$

where $\Delta_1 = \frac{1}{T} \sum_{t=1}^T \sum_{i \in \mathcal{B}} p_i^2 G^2 + \kappa^2$, $\Delta_2 = \frac{1}{T} \sum_{t=1}^T \sum_{i \in \mathcal{B}} p_i^2 (3\sigma^2 + \frac{3}{2}G^2) + \kappa^2$, w_1 and w^* are the initial and optimal global model respectively, η is the global learning rate, p_i is the weight of client i , and $\mathcal{B}$ is the cluster of the benign gradients selected by FedID in t round.

We observe that the global model converges as the number of rounds increases, eventually reaching the optimal solution after T rounds. While a larger η can accelerate convergence due to its presence in the denominator, it may also cause instability and divergence, as reflected in the second term. When the client's local data are IID and the full gradient is calculated, the convergence errors caused by σ^2 and G^2 will be eliminated. Conversely, convergence performance will deteriorate as non-IID degrees increase. The malicious impact κ^2 also negatively degrades the convergence performance. As we find that $\kappa^2 \propto |\mu_B - \mu_M|^2$, the malicious updates are more easily detected along with the increase in malicious impact κ^2 due to the increase of $|\mu_B - \mu_M|$. Notably, the backdoor

attacks have small κ^2 as they do not expect a negative impact on the main performance and the convergence of the global model due to their adversarial goals. For untargeted attacks, we consider that they behave with significantly deviated μ_M , and our approach can effectively identify them.

The detailed proofs are shown in Appendix C. Moreover, we also empirically present the convergence performance of FedID adequately in Section VII-E.

V. ADAPTIVE ATTACK TARGETED TO BREAK THE DEFENSE

In the event that adversaries discern the implementation of our proposed defensive strategy to mitigate their intrusions, there exists the potential for them to engineer specialized strategies to bypass our protective measures. In alignment with Kerckhoffs's Principle, we operate under the presumption that the adversary has a comprehensive understanding of our delineated defense mechanism and thus, is capable of orchestrating an adaptive assault specifically tailored to subvert it. In this section, we provide a detailed investigation of possible attacks targeted to break our defense approach.

Since various defense mechanisms rely on distance metrics to detect backdoor gradients, prior work [9] proposes to modify the loss function to ensure that it will not be rejected by the anomaly detector employed at the server by adding an anomaly detection loss term $\mathcal{L}_{\text{ano}}$ to adjust the characteristics of the backdoor gradient. The formulation of the augmented loss function can be articulated as:

$$\mathcal{L}_{\text{model}} = \beta \mathcal{L}_{\text{class}} + (1 - \beta) \mathcal{L}_{\text{ano}}, \quad (8)$$

where $\mathcal{L}_{\text{class}}$ encapsulates the accuracy metrics pertaining to both benign and backdoor instances, while $\mathcal{L}_{\text{ano}}$ is the anomaly detection loss term which ensures that the malicious gradients bypass the defense. The parameter β controls the trade-off between maintaining the attack's effectiveness and circumventing the defense mechanism. Generally, $\mathcal{L}_{\text{ano}}$ quantifies the divergence of the attack model from the benign model in terms of the metrics used by the defense. For example, if the defense employs Euclidean distance to differentiate backdoor gradients and the benign ones, the attacker would set $\mathcal{L}_{\text{ano}}$ as the Euclidean distance between the attack model (i.e., the model update sent to the server) and the last received global model.

We present FedID, which employs three distance metrics to capture the diverse characteristics of different attacks. Attackers, in the event that they are aware of our defense, will also attempt to obfuscate the backdoor gradient by taking all three metrics into consideration to bypass our defense. Consequently, they will introduce the anomaly loss of these three metrics into the model loss function as follows:

$$\mathcal{L}_{\text{ano}} = \mathcal{L}_{\text{Man}} + \mathcal{L}_{\text{Eul}} + \mathcal{L}_{\text{Cos}}. \quad (9)$$

where $\mathcal{L}_{\text{Man}}$, $\mathcal{L}_{\text{Eul}}$ and $\mathcal{L}_{\text{Cos}}$ represents the Manhattan, Euclidean, and Cosine distance between the attack model and the benign model on the attacker's dataset. We provide the specifically designed attack in Algorithm 2. We argue that our defense has the capacity to safeguard the FL system even when

TABLE I
THE DEFAULT FL SYSTEM PARAMETER SETTINGS.

Backdoor type		Semantic trigger					Artificial trigger			
$\mathcal{D}$	dataset	CIFAR10	CINIC10	EMNIST	Fashion-MNIST	Sentiment140	CIFAR10	EMNIST	Fashion-MNIST	LOAN
$\mathcal{D}_A$	backdoor data	Southwest Airlines		Ardis		Greek Director	-			
N	#clients	200				200	300	100		
K	#clients selected in each round	10								
-	#attackers in each round	1					4			
-	#attackers local iteration	2				5	6	10		5
-	#benign local iteration	2						1		
T	#global iteration	1500			500	300	70			
-	batch size	32				20	64			
-	attack interval	10					1			
α	non-IID parameter	0.5				1	0.5			0.9
η	benign learning rate	0.02				0.05	0.1			0.001
η_A	attackers learning rate	0.02				0.05				0.0005

Algorithm 2 Adaptive attack targeted to our defense

Input: Initial global from the server w_0 , attacker's dataset $\mathcal{D}_A$, local learning rate lr_{adv} , local training epoch E

Output: Attack model w_{adv}

```

1:  $w_A \leftarrow w_0$ 
2: for  $i \in \{1, 2, \dots, E\}$  do
3:    $\mathcal{L}_{Man} \leftarrow distance_{Man}(X, w_0)$   $\triangleright$  compute the
     distance Manhattan distance
4:    $\mathcal{L}_{Eul} \leftarrow distance_{Eul}(X, w_0)$   $\triangleright$  compute the distance
     Euclidean distance
5:    $\mathcal{L}_{Cos} \leftarrow distance_{Cos}(X, w_0)$   $\triangleright$  compute the distance
     cosine distance
6:    $\mathcal{L}_{ano} \leftarrow \mathcal{L}_{Man} + \mathcal{L}_{Eul} + \mathcal{L}_{Cos}$ 
7:    $\mathcal{L}_{model} \leftarrow \beta \cdot \mathcal{L}_{class} + (1 - \beta) \cdot \mathcal{L}_{ano}$ 
8:    $w_A \leftarrow w_A - lr_{adv} \cdot \nabla \mathcal{L}_{model}(w_A, \mathcal{D}_A)$   $\triangleright$  gradient
     descent
9: end for
10:  $w_{adv} \leftarrow w_A$ 

```

attackers employ the adaptive attack with varying β values, as demonstrated in Section VII-C.

VI. EXPERIMENT SETUP

We will first introduce the settings and then the objective of our experiments.

Datasets. We conduct our experiments on multiple datasets, including three visual datasets, CIFAR-10 [57] EMNIST [58], and Fashion-MNIST [59]. We also experiment on other datasets to prove the generalization ability, including CINIC10 [60], LOAN [61] and Sentiment140 [62] as follows:

- **CIFAR-10 dataset [57]** is a vision classification dataset containing 50,000 training samples and 10,000 testing samples with ten classes. To simulate the FL environment, we set the α of the Dirichlet distribution to be 0.5, representing a non-IID distribution. Smaller α means a larger degree of non-IID.
- **EMNIST dataset [58]** is a digit classification dataset consisting of 280,000 real-world handwritten images of digits from 0 to 9. We also set the degree of non-IID to be $\alpha = 0.5$.
- **Fashion-MNIST dataset [59]** is a dataset comprising 70,000 fashion products from 10 categories, with 7,000

images per category. Similarly, we set the non-IID parameter $\alpha = 0.5$.

- **CINIC10 [60]** has a total of 270,000 images, 4.5 times that of CIFAR-10, which is constructed from two different sources: ImageNet and CIFAR-10. We set the non-IID degree as $\alpha = 0.5$.
- **Lending Club Loan dataset (LOAN) [61]** contains financial information such as loan information and credit scores for loan status prediction. There are 2,260,668 samples with a total of 9 classes. To simulate a more realistic situation, we divide the data by US states, where each state represents one client. We set the non-IID parameter $\alpha = 0.9$.
- **Sentiment140 [62]** is a publicly available English Twitter sentiment classification dataset containing 1.6 million tweets, of which 800,000 are for training and the other 800,000 are for testing. Each tweet in the dataset is labeled with a sentiment polarity (positive, neutral, or negative).

Models. Different models are used on different datasets to demonstrate the generality of our approach. A 9-layer VGG style network (VGG-9) [63] is trained on the CIFAR-10. On the EMNIST, we train LeNet-5 model [64]. We train the LSTM model [65] for Sentiment140 and a network with three fully-connected layers for LOAN.

Baseline Defenses. We implement eight representative FL aggregation and backdoor defense algorithms by their source code as baselines below.

- **FedAvg(no defense) [2]:** Aggregates all gradients purely without any defense.
- **Krum [21]:** Selects a gradient with the smallest Euclidean distances to the others as the global update this round.
- **Multi-Krum [21]:** Excludes several gradients with the largest Euclidean distances to the others and aggregates the remains, which is an extended version of Krum.
- **RFA [23]:** Aggregates the local gradients by computing the geometric median.
- **Foolsgold [20]:** Identifies malicious gradients by their historical cosine similarity.
- **Weak-DP [31]:** Clips the norm of gradients and adds the Gaussian noise to the global model for mitigating the backdoors. The default standard deviation of Gaussian noise is 0.0025, and the clipping bound is 2.

- **FLPruning** [66], which find the backdoor neurons after voting by clients and prune them based on a validated dataset.
- **Flame** [19], which adopts a clustering method using HDBSCAN and cosine similarity in addition to the DP process (clipping and nosing) and only adds minimum noise (noise $\sigma = 0.001$) to the model.

Backdoor Attacks. We implement both pixel-pattern and semantic backdoors. To make the attack more challenging, we use the multiple-shot attack instead of the single-shot. The following are the attack methods we use:

- **Model Replacement Attack** [9]: Malicious clients generate backdoor gradients during their local training. Attackers scale up their gradients to ensure that the malicious model will replace the global model. Usually, the scaling factor is N/K ;
- **DBA Attack** [12]: DBA decomposes a global trigger pattern into separate local patterns and embeds them into the training sets of different attackers, resulting in a more negligible difference between benign and backdoor gradients;
- **PGD Attack** [11]: To bypass the Euclidean-based defense, PGD Attack projects the model parameter within the ball centered around the global model;
- **Edge-case PGD Attack** [11]: Attackers have a set of edge-case samples and some benign samples as their poisoned datasets. Edge-case samples are unlikely to exist in the benign training set. Thus, the benign gradients cannot dilute the backdoor. We launch this attack in the form of PGD. The Edge-case PGD attack is so stealthy that most defenses cannot defend against it.

Evaluation. We leverage the following two metrics for evaluating the effectiveness of our defense:

- **Backdoor Accuracy (BA)** indicates whether the attacker succeeds. It shows the accuracy of the model on the backdoor task. The attackers aim to maximize this metric, and the lower the BA, the more efficient the defense.
- **Main Task Accuracy (MA)** is the accuracy on the main task. The overall objective of all participants in an FL system is to maximize MA. The benign clients maximize the model performance, while the attackers seek to be stealthy and maintain MA. Thus, the defense should not cause a significant drop in MA, which we report to demonstrate that our defense does not affect the functionality of the FL system. We report all BA and MA values in percentages.

Hyperparameters We implement the attacks following the source code provided by Bagdasaryan et al. [9], Xie et al. [12], and Wang et al. [11], and detailed the default FL system parameters in Table I.

Objectives of Experiments. Firstly, we aim to illustrate the effectiveness of our proposed defense under different attacks by comparing it with the current and previous state-of-the-art defenses. Secondly, we demonstrate that the performance of our defense is consistent under different scenarios (e.g., different degrees of non-IID α and different attack frequencies). Thirdly, we evaluate that our proposed method achieves

efficient performance against the potential adaptive attack and untargeted attacks. Finally, we figure out the reason for our defense works through the ablation study in Section VII-F.

VII. EXPERIMENT RESULTS

A. Robustness against Different Attacks and Comparison with SOTA

We first demonstrate the robustness of FedID under different attacks and then compare our methods with the state-of-the-art defense methods such as Krum [21], Multi-Krum [21], Foolsgold [20], RFA [23], Weak-DP [31], FLpruning [66] and Flame [19] in Table II. All the defenses except RFA and Weak-DP successfully defended against the model replacement attack since RFA and Weak-DP still incorporate some malicious gradient into the global model. FedID achieves the lowest BA on both datasets (0.56% and 0.00%) with little impact on the MA. Under the DBA attack, Multi-Krum almost has no effect. Flame and our method also deteriorate because 40% of the participants in each round are malicious, which significantly impacts the method of selecting multiple clients for aggregation. Although Krum and Foolsgold successfully defend against the DBA attack, they also negatively affect the overall performance of the global model. Our defense can reduce the BA to less than 10% without affecting the MA.

PGD attack is much stealthier than the above two, which Krum, RFA, and Multi-Krum fail to detect. Moreover, once they select the wrong gradient for aggregation, the global model is replaced, leading to a worse BA than FedAvg. FedID and Flame achieve the best effect of 0.56% and 0.00% on the two datasets, respectively. Edge-case PGD is the hardest of all since it is both stealthy and effective since it attacks with edge-case samples and projects the gradient back to the L_2 norm ball. All the previous defenses have limited effect on this attack. Since backdoor neurons and benign neurons are not completely independent of each other, FLpruning cannot eliminate all backdoors, causing its failure against this attack. Flame reduces the BA to 5.12%. However, Flame suffers from a 6% drop on MA, which is a result of adding noise. In comparison, FedID achieves a much lower BA of 3.06% with almost no impact on MA.

On EMNIST, we also achieve better defense compared to up-to-date methods. Remarkably, we achieve 0.0% BA under both PGD and Edge-case PGD attacks meaning that we defend entirely against these two adversaries. The proposed method also achieves state-of-the-art performance compared to other defenses on Fashion-MNIST while maintaining a high MA and low BA. We also achieve similar performance without sacrificing the benign performance against model replacement and DBA. To provide an overall comparison with previous methods, we follow [67] and design an improved version of the ranking score for each defense with respect to baseline FedAvg. We use the percentage of improvement over the baseline to score and compute it on MA/BA respectively, formally as $\frac{K_{MA/BA} - B_{MA/BA}}{B_{MA/BA}}$ where K denotes some methods and B denotes baseline. Adding up $(score_{MA} - score_{BA})$ across all attacks produces the ranking score for each defense. As shown in the last column of Table II, we show that our

TABLE II
ROBUSTNESS OF FEDID COMPARED TO THE SOTA DEFENSES FOR VARIOUS CHALLENGING ATTACKS.

Dataset	Defense	Model Replacement		DBA		PGD		Edge-case PGD		Rank Score
		MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	
CIFAR10	FedAvg	86.95	64.80	79.23	90.44	87.04	14.44	87.14	55.10	+0.00
	RFA	86.69(+0.00)	25.56(-0.61)	79.6(+0.00)	57.69(-0.36)	87.10(+0.00)	52.56(+2.64)	86.47(-0.01)	65.31(+0.19)	-1.86
	Foolsgold	85.71(-0.01)	6.67(-0.90)	77.56(-0.02)	3.43(-0.96)	84.92(-0.02)	14.44(+0.00)	84.76(-0.03)	51.53(-0.06)	+1.84
	Krum	82.17(-0.05)	6.11(-0.91)	78.18(-0.01)	6.01(-0.93)	82.32(-0.05)	66.67(+3.62)	81.23(-0.07)	59.18(+0.07)	-2.04
	Multi-Krum	86.55(+0.00)	1.67(-0.97)	79.33(+0.00)	91.39(+0.01)	86.52(-0.01)	17.78(+0.23)	87.4(+0.00)	60.20(+0.09)	+0.63
	Weak-DP	74.41(-0.14)	46.11(-0.29)	N/A	N/A	74.43(-0.14)	22.22(+0.54)	73.84(-0.15)	53.06(-0.04)	-0.66
	FLpruning	85.72(-0.01)	6.67(-0.90)	78.23(-0.01)	8.57(-0.91)	85.77(-0.01)	9.44(-0.35)	85.74(-0.02)	48.47(-0.12)	+2.23
	Flame	80.58(-0.07)	0.56(-0.99)	76.78(-0.03)	37.24(-0.59)	81.24(-0.07)	0.56(-0.96)	81.41(-0.07)	5.12(-0.91)	+3.21
	FedID	86.66(+0.00)	0.56(-0.99)	79.46(+0.00)	7.24(-0.92)	86.47(-0.01)	0.56(-0.96)	86.58(-0.01)	2.04(-0.96)	+3.82
EMNIST	FedAvg	99.54	96.00	97.68	94.13	99.55	10.00	99.37	96.00	+0.00
	RFA	99.57(+0.00)	6.00(-0.94)	97.87(+0.00)	1.39(-0.99)	99.32(+0.00)	4.00(-0.60)	99.29(+0.00)	97.00(+0.01)	+2.51
	Foolsgold	96.42(-0.03)	98.00(+0.02)	97.24(+0.00)	0.64(-0.99)	99.07(+0.00)	94.00(+8.40)	99.13(+0.00)	98.00(+0.02)	-7.49
	Krum	99.22(+0.00)	0.00(-1.00)	97.7(+0.00)	0.56(-0.99)	99.12(+0.00)	1.00(-0.90)	99.14(+0.00)	12.00(-0.88)	+3.76
	Multi-Krum	99.58(+0.00)	0.00(-1.00)	97.85(+0.00)	47.43(-0.50)	99.54(+0.00)	0.00(-1.00)	99.57(+0.00)	84.00(-0.13)	+2.63
	Weak-DP	99.37(+0.00)	86.00(-0.10)	N/A	N/A	99.41(+0.00)	14.00(+0.40)	99.39(+0.00)	89.00(-0.07)	-0.23
	FLpruning	99.41(+0.00)	24.00(-0.75)	97.24(+0.00)	4.79(-0.95)	99.48(+0.00)	14.00(+0.40)	99.46(+0.00)	92.00(-0.04)	+2.14
	Flame	99.39(+0.00)	0.00(-1.00)	97.12(-0.01)	17.38(-0.82)	99.39(+0.00)	0.00(-1.00)	99.44(+0.00)	13.00(-0.86)	+3.67
	FedID	99.49(+0.00)	0.00(-1.00)	97.22(+0.00)	3.76(-0.96)	99.57(+0.00)	0.00(-1.00)	99.46(+0.00)	0.00(-1.00)	+3.96
Fashion-MNIST	FedAvg	91.50	99.00	87.59	99.69	91.60	99.00	91.95	99.00	+0.00
	RFA	89.76(-0.02)	99.00(+0.00)	87.41(-0.06)	44.21(-0.56)	91.20(+0.00)	99.00(+0.00)	91.04(-0.01)	99.00(+0.00)	+0.52
	Foolsgold	87.62(-0.04)	98.00(-0.01)	82.58(-0.05)	92.78(-0.07)	87.11(-0.05)	35.00(-0.65)	90.74(-0.01)	99.00(-1.00)	+0.56
	Krum	87.21(-0.05)	0.00(-1.00)	82.87(-0.05)	8.57(-0.91)	86.27(-0.06)	0.00(-1.00)	85.05(-0.08)	0.00(-1.00)	+3.68
	Multi-Krum	90.92(-0.01)	0.00(-1.00)	87.19(+0.00)	3.08(-0.97)	90.98(-0.01)	1.00(-0.99)	90.08(-0.02)	97.00(-0.02)	+2.94
	Weak-DP	90.51(-0.01)	0.00(-1.00)	87.11(-0.01)	99.81(+0.00)	89.92(-0.02)	17.00(-0.83)	89.34(-0.03)	99.00(+0.00)	+1.89
	FLpruning	N/A	N/A	85.63(-0.03)	9.81(-0.90)	88.38(-0.04)	15.00(-0.85)	88.68(-0.04)	87.00(-0.12)	+1.66
	Flame	89.12(-0.03)	0.00(-1.00)	86.26(-0.02)	3.26(-0.97)	88.24(-0.04)	0.00(-1.00)	87.16(-0.05)	0.00(-1.00)	+3.84
	FedID	90.32(-0.01)	0.00(-1.00)	86.33(-0.01)	2.81(-0.97)	90.52(-0.01)	0.00(-1.00)	89.80(-0.02)	4.00(-0.96)	+3.87

method obtains the highest ranking score with almost 400% better than the baseline and outperforms the second-best Flame by around 0.5. We detail the ranking score in Appendix D.

still much better than the compared defenses, e.g., Krum and Foolsgold.

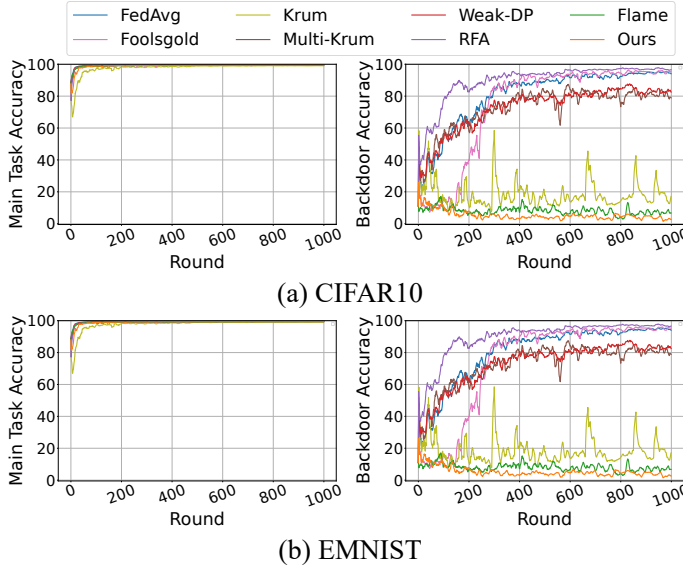


Fig. 4. MA(%) and BA(%) of various defenses under Edge-case PGD attack.

We show the MA and BA along the training process of the seven defenses under the Edge-case PGD attack in Figure 4. Only our method and Flame successfully resist the attack during the entire training process, and Flame also dampens the MA. The disadvantage of our approach is that the convergence speed is slightly slowed down compared to FedAvg, but it is

B. Impact of Training Environment

In this section, we demonstrate that our defense is invariant to the different training environments, i.e., degrees of non-IID and attack frequency.

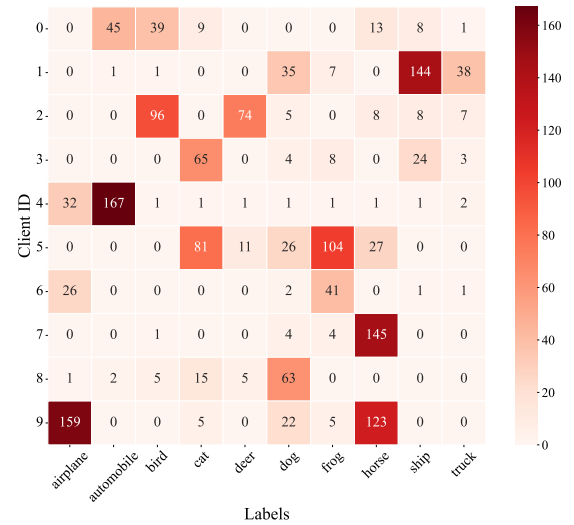


Fig. 5. Label distribution on different clients under $\alpha = 0.1$.

Impact of Different Degrees of Non-IID. As mentioned abundantly in Section III-B, the different data distributions contribute to one of the hardest problems in distance-based

defense. Thus, to illustrate the effectiveness of our method, we demonstrate that our defense is invariant to the different degrees of non-IID. The experiment is conducted with the Edge-case PGD attack. We simulate different degrees of non-IID by adjusting the value of parameter α . As shown in Figure 5, the label distribution on different clients in a random round under $\alpha = 0.1$. We find that the significant heterogeneity stems from substantial differences in the total data amount and that each client mainly possesses data from a few classes. Figure 6 shows that the BA remains low in all cases. Our method only slightly impacts the MA with $\alpha < 0.2$, while maintaining BA close to 0.

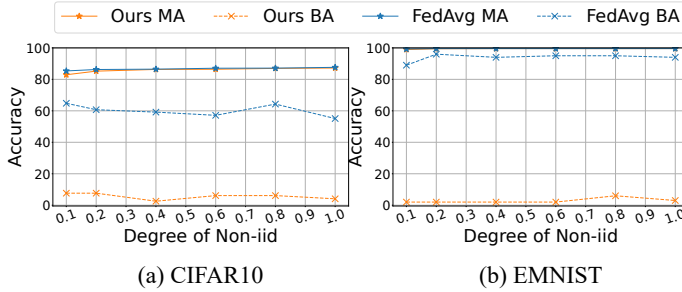


Fig. 6. Accuracy(%) of our defense under Edge-case PGD attack on different degrees of non-IID, which is indicated by the α parameter of the Dirichlet distribution.

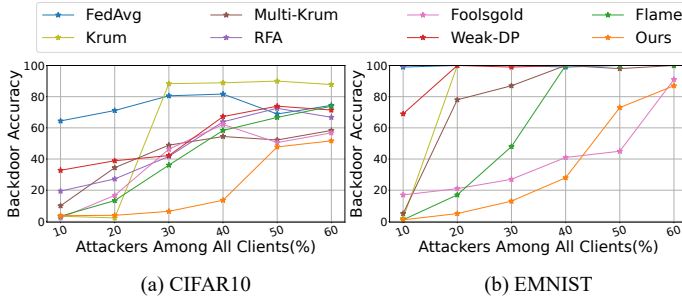


Fig. 7. BA(%) of various defenses under the backdoor attack versus percentage of attackers among all clients.

Impact of Attacker Percentage. We show the BA performance with different ratios of attackers under the backdoor attack in Figure 7, where we compare FedID with other methods. As a result of our rigorous experiment settings, it usually selects more than $K/2$ adversarial models in each round, so all defenses, especially the clustering-based methods, fail to defend when 50% of clients are compromised. For EMNIST, BA rises steeply as the simplicity of the task, achieving high accuracy with few parameters injected. Nonetheless, our method performs significantly superior to other methods and exhibits invariance to the number of attackers encountered.

Impact of Attack Frequency. The frequency of the attack affects the strength and persistence of the attack. We demonstrate that our defense is invariant to the frequency of attack in Figure 8, where we show that our defense achieves consistently low BA under all attack intervals. Since the other defenses cannot resist the edge-case PGD with the minimum frequency (attack interval of 10), we do not plot their results.

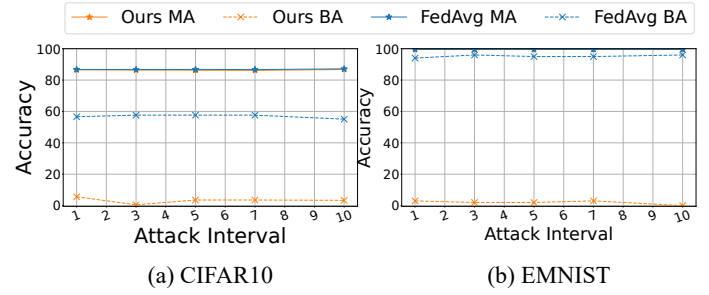


Fig. 8. Accuracy(%) of our defense under Edge-case PGD attack versus attack interval.

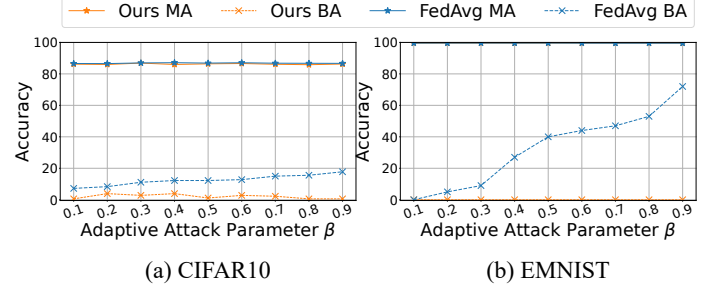


Fig. 9. Accuracy(%) versus parameter β graph of various defenses against the adaptive attack designed to break our defense.

C. Robustness against Targeted Adaptive Attack Tailored to Break our Defense

We demonstrate the effectiveness and generalizable performance of our defense by showing that our approach successfully defends against this potential adaptive attack specifically designed to break our system. In examining Figure 9, it becomes discernible that for values of β less than 0.3, the adaptive attack is predominantly oriented towards obfuscating malicious intent and thus provides a low backdoor accuracy. However, the backdoor accuracy (BA) against FedAvg experiences rapid growth as β exceeds 0.3, particularly on the EMNIST dataset while our approach consistently maintains a low BA close to zero and keeping the model accuracy (MA) at the same level as FedAvg. Consequently, we can reach the conclusion that the adaptive attack will either be detected by our defense or remain concealed, ultimately losing its threat to the FL system.

We set β to 0.5 and assess the effectiveness of FedID against the potential adaptive attack, including A LITTLE IS ENOUGH [68], RLBackdoor [69], and the adaptive attack tailored to FedID mentioned in Section V, as presented in Table III. A LITTLE IS ENOUGH clips its attack gradient around the global model to achieve stealthiness. RLBackdoor is the backdoor version of [70], which attacks FL based on reinforcement learning. Here, we leverage edge-case data to conduct these attacks to amplify challenging.

The defenses perform similarly against A LITTLE IS ENOUGH and the PGD attack, as these two attacks are based on the same mechanism that clips or projects the attack model around the global model to achieve stealthiness. The results show that FedID is effective in detecting these

TABLE III
EFFECTIVENESS OF VARIOUS DEFENSES AGAINST THE ADAPTIVE ATTACK.

Datasets	Defenses	LITTLE IS ENOUGH		RLBackdoor		Adaptive	
		MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$
CIFAR10	FedAvg	86.91	59.18	85.04	98.90	86.89	43.37
	RFA	86.85	33.67	84.98	99.10	86.78	51.02
	Foolsgold	85.09	75.51	84.02	4.50	85.06	26.02
	Krum	81.53	4.08	83.37	5.60	81.96	56.12
	Multi-Krum	86.75	7.14	84.64	5.50	86.53	43.37
	Weak-DP	72.58	47.96	74.64	98.50	73.64	44.39
	Flame	83.04	7.14	79.29	8.50	81.41	51.53
EMNIST	FedID	86.16	2.55	84.78	3.70	86.40	9.69
	FedAvg	99.54	80.00	95.38	97.75	99.51	92.00
	RFA	99.50	72.00	95.31	97.45	99.54	95.00
	Foolsgold	99.44	91.00	94.43	1.90	99.53	67.00
	Krum	98.64	11.00	94.23	1.80	99.11	12.00
	Multi-Krum	99.53	4.00	95.26	1.50	99.57	45.00
	Weak-DP	92.56	68.00	92.20	95.65	99.00	89.00
Fashion-MNIST	Flame	99.42	42.00	93.47	1.45	99.43	30.00
	FedID	99.45	4.00	95.06	1.10	99.57	6.00
	FedAvg	90.83	99.00	85.28	99.20	92.11	99.00
	RFA	90.24	99.00	84.81	98.80	91.80	97.00
	Foolsgold	90.32	99.00	83.87	0.00	91.34	48.00
	Krum	87.08	0.00	82.38	0.00	87.13	0.00
	Multi-Krum	90.75	0.00	85.11	0.00	90.20	98.00
Fashion-MNIST	Weak-DP	88.77	98.00	81.40	97.70	90.32	99.00
	Flame	89.13	0.00	83.09	0.00	89.83	95.00
	FedID	89.88	0.00	84.95	0.00	89.49	8.00

TABLE IV
EFFECTIVENESS OF VARIOUS DEFENSES AGAINST UNTARGETED ATTACK.

Defenses	Sign-flipping		Gaussian		MODEL POISONING	
	CIFAR10	EMNIST	CIFAR10	EMNIST	CIFAR10	EMNIST
FedAvg	N/A	45.41	N/A	96.98	N/A	N/A
RFA	86.78	98.78	86.64	99.59	N/A	N/A
Krum	81.93	99.17	81.93	99.12	N/A	N/A
Multi-Krum	86.55	99.54	86.55	99.56	86.29	99.52
Weak-DP	65.67	96.90	68.13	99.28	N/A	N/A
Flame	64.28	98.25	81.30	99.41	81.97	98.66
FedID	86.36	99.56	86.05	99.54	86.14	99.47

attacks. Although RLBackdoor is so effective that it achieves a BA of 98.90% against FedAvg on CIFAR10, it sacrifices its stealthiness and even fails against Foolsgold. The adaptive attack successfully breaches nearly all of the defense mechanisms. This suggests that the adaptive attack is considerably covert that existing defenses cannot detect it. However, our proposed method consistently performs effectively across all three datasets. This further demonstrates that our defense performance can be generalized to different attack settings.

D. Robustness against Untargeted Attacks

We conducted three untargeted attacks, namely the Sign-flipping attack, Gaussian attack [17], and MODEL POISONING attack [71] to evaluate the robustness of various defenses, shown in Table IV. The objective of untargeted attacks is to degrade the overall model performance, potentially leading to the failure of the model training. Malicious updates in untargeted attacks are often more conspicuous compared to targeted attacks(i.e., backdoor attacks). Consequently, distance-based defenses like Krum can effectively resist untargeted attacks because distinguishing unconcealed attacks is relatively straightforward for them. On the other hand, defenses relying

on the differential privacy (DP) method may not entirely thwart untargeted attacks, but their clipping process provides some level of protection. However, the advanced MODEL POISONING attack successfully circumvents the detection and completely destroys the model, except for Multi-Krum and FedID. Our approach successfully defends against these three untargeted attacks on CIFAR10 and EMNIST datasets, demonstrating its robustness in the face of untargeted attacks.

E. Convergence and Generalization Performance to Different Datasets

Firstly, we evaluate the convergence performance of FedID on different datasets compared to FedAvg, as shown in Figure 10. The results illustrate that the accuracy of FedID is close to FedAvg after 1/3 of training rounds on CIFAR10 and CINIC10. On datasets with simpler features, EMNIST and Sentiment140, FedID performs similarly to FedAvg throughout the training process. These results demonstrate a good convergence performance of FedID.

Secondly, we show the backdoor and main accuracy of FedID on the CINIC10, LOAN, and Sentiment140 in Table V. Our method achieves high MA and low BA on both datasets, proving the generalization to different models and tasks.

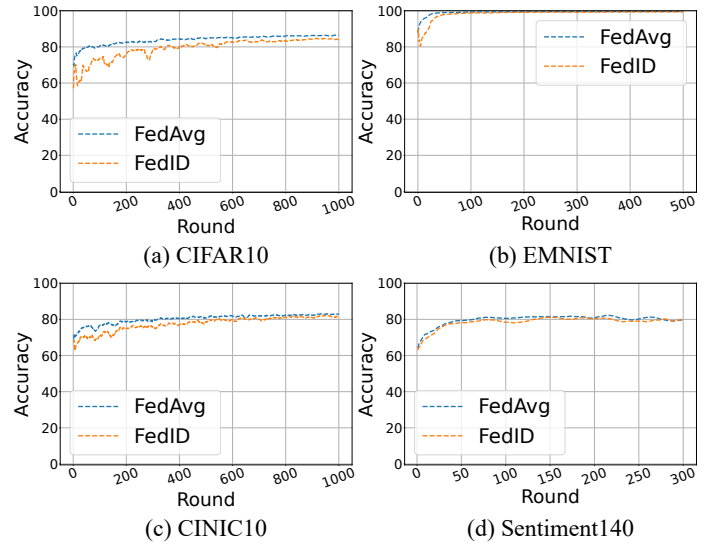


Fig. 10. Convergence performance of FedID on different datasets.

TABLE V
EFFECTIVENESS OF OUR METHOD AGAINST BACKDOOR ATTACK ON OTHER DATASETS.

Defense	CINIC10		LOAN		Sentiment140	
	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$
FedAvg	84.37	36.22	89.05	61.36	82.59	89.17
FedID	83.02	4.59	88.52	0.00	81.67	5.83

F. Ablation Study

The above experiments present the remarkable defense ability of our approach. In this section, we conduct an ablation study to understand the role of each component.

TABLE VI
EFFECTIVENESS OF THE METRICS IN OUR APPROACH TO DEFENDING AGAINST THE VARIOUS ATTACKS ON CIFAR10.

Defense	Model Replacement	PGD	Edge-case PGD
	MA/BA	MA/BA	MA/BA
Man	83.86/ 0.56	83.74/25.56	85.3/64.80
Eul	86.24/ 0.56	85.52/17.78	87.12 /54.08
Cos	84.22/2.22	83.84/30.00	85.38/66.84
Man+Eul	84.09/1.11	84.17/28.63	84.3/67.35
Man+Cosine	85.74/1.67	85.16/23.68	85.86/6.63
Cosine+Eul	86.31/ 0.56	85.44/16.11	85.14/63.78
Man+Cosine+Eul	86.66 / 0.56	86.47 / 0.56	86.58/ 2.04

Multi-metrics Identifies the Stealthiest Attacks. We first conduct an ablation study on the multi-metrics, as shown in Table VI. We observe the following: 1. All three metrics perform poorly individually. 2. Manhattan distance improves the defense ability when combined with the other metrics. 3. Cosine similarity combined with Manhattan distance provides the best defense against the Edge-case PGD. Given the above observations, we conclude that multi-metrics adaptively defends against universal backdoors. The introduced Manhattan metric contributes to the defense against the stealthiest attack (e.g., Edge-case PGD). To further demonstrate the claim that a single metric cannot identify malicious gradients of diverse characteristics, we show the dominance of each metric under different attacks. We take the metric as the dominant one with the most considerable contribution to this round's scoring and plot their frequency in Figure 11.

Firstly, we show that Manhattan distance contributes to the defense of PGD and Edge-case PGD, complementing the Cosine similarity justifying the above observations 2 and 3. We also notice that Euclidean is never dominant under PGD and Edge-case PGD, consistent with the analysis that these two attacks bypass L_2 norm through projection back to L_2 norm ball. We also notice that Manhattan distance plays no role under the model replacement attack, consistent with the above experiment that Manhattan does not affect the final BA. It is because the model replacement attack scales the gradient of the malicious client, making it susceptible to cosine and Euclidean distances. Finally, we conclude that better defense can be achieved if multiple instead of single metrics are utilized since different metrics dominantly detect different malicious gradients.

Dynamic Weighting through Whitening Process Adaptively Adjust the Scoring. We illustrate the effectiveness of our whitening process discussed in Section III-B in Table VII by comparing it with the standard max normalization. BA increases by a large margin if only max normalization is applied, which illustrates that the whitening process plays an essential role in identifying the malicious under non-IID distributions. Moreover, we notice the MA increases as we apply the whitening process, which shows that it can flexibly identify the benign but different gradients (due to non-IID) and the true malicious ones.

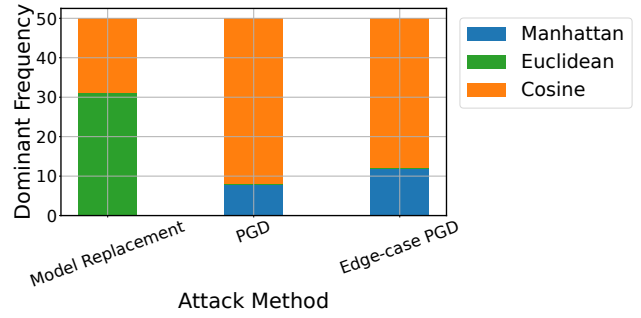


Fig. 11. Frequency of dominance of each metric under different attacks on CIFAR10.

TABLE VII
EFFECTIVENESS OF DIFFERENT WEIGHTING METHODS ON DEFENSE AGAINST VARIOUS ATTACKS ON CIFAR10.

Weighting	Model Replacement		PGD		Edge-case PGD	
	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$
Max Norm	83.86	0.56	83.74	25.56	84.08	62.24
Whitening	86.34	0.56	86.66	0.56	86.58	2.04

G. Scalability of the Method and Computational Cost Analysis

In Section III-B, we introduce a dynamic weighting approach that enhances the scalability of FedID. As illustrated in Table VIII, we present the results of FedID under various additional distance metrics, including Jaccard distance, Bray-Curtis distance, and Canberra distance. Our findings reveal that the absence of dynamic weighting leads to two significant challenges: 1) The use of ineffective metrics, such as the Jaccard distance, fails to contribute positively to the defense mechanism; 2) The incorporation of multiple metrics negatively impacts the main task performance of the FL system. The dynamic weighting mechanism in FedID enables the system to adapt to diverse scenarios by effectively leveraging relevant metrics. With an increasing number of metrics, FedID experiences substantial improvements in effectiveness and robustness.

Nevertheless, including additional metrics in the methodology (i.e., FedID+Jacc+Bray+Canb) comes with an associated increase in computational costs. As demonstrated in Figure 12, this heightened computational overhead primarily arises from the computation of extra distance metrics between models, with the Bray-Curtis distance and the Canberra distance being particularly resource-intensive. A substantial portion can also

TABLE VIII
EFFECTIVENESS OF DYNAMIC WEIGHTING COMBINED WITH MORE METRICS AGAINST EDGE-CASE PGD ATTACK ON CIFAR10.

Defenses	w/. Dynamic Weight		w/o. Dynamic Weight	
	MA $\uparrow$	BA $\downarrow$	MA $\uparrow$	BA $\downarrow$
FedID+Jacc	86.23	3.06	84.12	56.63
FedID+Bray	85.37	2.55	83.79	5.10
FedID+Canb	86.24	4.59	83.57	4.08
FedID+Jacc+Bray	87.19	2.55	83.23	4.08
FedID+Jacc+Bray+Canb	86.50	1.02	83.81	3.57

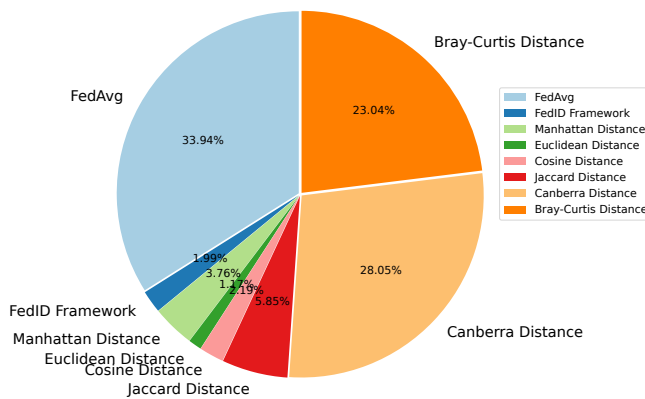


Fig. 12. Percentage of computational cost for each component of the method.

be attributed to the FedAvg, primarily due to the local training across selected clients in each round. Notably, when examining the basic FedID (i.e., the FedID Framework utilizing Manhattan, Euclidean, and Cosine distances), it is evident that it accounts for only 9.11% of the total computational cost.

VIII. CONCLUSION

Existing defense strategies often fall short when confronted with the intricate challenge of backdoor attacks, where malicious updates closely mimic benign data. This vulnerability is exacerbated when benign clients exhibit non-IID distribution data. In response, we introduce FedID, a novel and dynamic defense mechanism. By employing multiple metrics with dynamic weighting and integrating a modified z-score approach for update aggregation, FedID showcases remarkable resilience against a spectrum of attack scenarios. Besides, we substantiate both theoretically and empirically that the Manhattan distance metric significantly outperforms the Euclidean distance, mitigating the issue of the “curse of dimensionality”. Most significantly, FedID operates independently of predefined assumptions, which is often a limiting factor in many existing defense strategies. In addition, it imposes only minimal impacts on benign performance. Moreover, through rigorous evaluations across diverse attack scenarios, we have consistently demonstrated the efficacy of FedID. Even when confronted with potential adaptive attacks specifically designed to subvert its defenses, FedID continues to shine as a state-of-the-art solution.

IX. ACKNOWLEDGEMENTS

This work is supported by the National Natural Science Foundation of China (Grant No. 62476095).

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